

Presentation and challenge session

Project: Reproducible AI Research Study

1. Your mission

Your learning target

I can explain the mechanism and terminology of presentation and challenge session.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Final project manuscript. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Peer review and response. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Figures that preserve evidence. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of presentation and challenge session

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Final project manuscript. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

My starting explanation

This week's end goal

Explain presentation and challenge session using the records and boundaries of this fictional investigation.

2. Notice the evidence

A presentation should make the evidence easy to challenge. Show the baseline, scope, negative cases and uncertainties instead of only a best example.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

List the given facts and one fact that is not supplied.

3. See how it works

Use a short sequence: question, design, primary result, failure, limitation, next question. Answer unfamiliar challenges by locating evidence or stating it is missing.

Presentation and challenge session — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: A presenter shows one perfect output after a 20-case study with 14 passes.

Check or calculation: Use a short sequence: question, design, primary result, failure, limitation, next question. Answer unfamiliar challenges by locating evidence or stating it is missing.

Conclusion and limit: The demonstration is one example; study rate 70%. Present aggregate and failures and do not call the best output representative without evidence.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for

qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

A presenter shows one perfect output after a 20-case study with 14 passes. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: The reviewer shows one camera-B success.

Prediction: That is one demonstration, not an estimate of broad camera-B reliability.

Reason: One selected observation does not represent the full proposed condition.

5. Words that unlock the idea

Word · Meaning

Word: Challenge session

Meaning: Questions testing the basis of a presentation

Word: Representative example

Meaning: An example whose selection is justified for the stated purpose

Word: Evidence response

Meaning: An answer grounded in records or explicit absence

Use two terms to explain a step in this week's example.

6. Follow a worked example

A presenter shows one perfect output after a 20-case study with 14 passes.

The demonstration is one example; study rate 70%. Present aggregate and failures and do not call the best output representative without evidence.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Presentation and challenge session

Answer a challenge without inventing evidence. At the presentation, a reviewer asks whether the model works on a different camera.

The facts we will use

The project tested twelve photos from camera A only. Ten were correct; no camera B records exist.

Presentation and challenge session

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State measured result

Report 10/12 correct on camera A.



2. Address the challenge

Say camera B performance is not measured.



3. Propose a valid next test

Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

CHANGE ONE THING

The reviewer shows one camera-B success.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. State measured result

What happens and why: Report 10/12 correct on camera A.

Step: 2. Address the challenge

What happens and why: Say camera B performance is not measured.

Step: 3. Propose a valid next test

What happens and why: Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A challenge session tests whether the team knows its evidence boundary. An honest unknown plus a specific future test is stronger than an unsupported confident promise.

What this example does not establish

The original result and proposed study still need sampling and uncertainty discussion.

Read the labelled picture: Answer the camera-B challenge using only the twelve camera-A records. Why is one displayed camera-B success a demonstration rather than broad evidence of camera-B reliability?

Change one thing: The reviewer shows one camera-B success. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

A confident response should not replace missing evidence with speculation.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A confident response should not replace missing evidence with speculation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Presentation and challenge session”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Presentation and challenge session. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Answer the camera-B challenge using only the twelve camera-A records. Why is one displayed camera-B success a demonstration rather than broad evidence of camera-B reliability?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

A presenter shows one perfect output after a20-case study with14 passes. Show the intermediate steps and give the conclusion with its scope.

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: A confident response should not replace missing evidence with speculation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final evidence pack: Audience asks if result holds on mobile; only desktop was tested. Give an honest answer. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside

any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Presentation and challenge session — evidence trace

Input and scope · Check or calculation · Conclusion and limit

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Check or calculation: Use a short sequence: question, design, primary result, failure, limitation, next question. Answer unfamiliar challenges by locating evidence or stating it is missing.

Conclusion and limit: The demonstration is one example; study rate 70%. Present aggregate and failures and do not call the best output representative without evidence.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Apply a new case independently

Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Answer a challenge without inventing evidence. At the presentation, a reviewer asks whether the model works on a different camera.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of presentation and challenge session

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of presentation and challenge session

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.